

Emilia Ma

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EDUCATION

University of Toronto

Sept 2024 – Expected May 2029

B.Sc. in Computer Science — Dean's List Scholar

GPA: 3.76/4.0

- Coursework: Data Structures & Algorithms, Software Design, Human Computer Interaction, Systems Programming, Computer Systems & Digital Logic, Theory of Computation, Probability & Statistics

EXPERIENCE

Ontario Public Service | Backend Dev/Junior Technical Analyst

Sept 2026 – Present

- Incoming software development intern starting September 2026 in the Digital Solutions Branch.
- Contributing to Java backend development for a large scale modernization of Ontario's driver and vehicle service systems, migrating legacy government systems to a new digital platform.
- Working in a multi layered enterprise backend architecture, on core application/business logic layers.

French Tutor & English Volunteer

2021–2024

- Taught English remotely to children in China with an emphasis on real conversational practice.
- Tutored a local student in French, tailoring sessions to her learning style.

PROJECTS

Tempo | *React, TypeScript, Node.js, Express, MongoDB, Better Auth, Railway*

2026–Present

Full-stack calendar focused on making scheduling faster and more intuitive through streamlined interactions.

- Independently designed, developed, and deployed Tempo end to end, implementing account authentication and user-scoped event storage so users can access their own calendars across sessions and devices.
- Engineered a regex parser that outperforms Google and Apple Calendar across 3+ tested scheduling scenarios, recognizing multi-day and combined date-time ranges (with “now” endpoints) in one field.
- Devised a one-click time entry workflow with instant AM/PM toggling for faster event creation.

Personal Portfolio | *React, TypeScript, HTML/CSS, Vite, GitHub Pages*

2026–Present

Interactive personal portfolio presented through a custom Mac-inspired, pixel-art browser interface.

- Created a portfolio featuring original pixel art, custom animations, and interactive React components.
- Reduced five large media assets by 94% (10.8 MB to 0.64 MB) through resizing and WebP conversion, significantly cutting initial load time from approximately 5–6 seconds to under 1 second.

GirlMath | *Python, FastAPI, React Native, TypeScript, MongoDB, DigitalOcean, Gemini API*

2026

AI powered journaling app supporting girls in STEM by tracking confidence, motivation, and learning trends.

- Led backend dev in a 36-hour hackathon, built FastAPI logic and REST endpoints for a 4 person team.
- Integrated the Gemini API to derive confidence, motivation, and learning trends from journal entries.
- Built frontend API client for backend communication and deployed the FastAPI service to DigitalOcean.

Munchables | *Java, Swing, OkHttp, JSON, Spoonacular API, HTML/CSS, JUnit*

2025

Four-person Java Swing social platform created with Clean Architecture and JUnit-tested domain logic.

- Designed a reusable Swing post component used across all post views, centralizing image rendering, comments, likes, and interaction logic.
- Integrated Spoonacular API, displaying nutrition data within the application using custom HTML/CSS.
- Implemented persistence via repository interfaces, isolating domain logic from storage infrastructure.

SKILLS

- **Programming Languages:** Python, Java, TypeScript/JavaScript, C, HTML/CSS, MIPS Assembly
- **Technologies & Tools:** Git, MongoDB, React/React Native, Node.js, Express, FastAPI, Maven, JUnit, Expo, Railway, DigitalOcean, Java Swing, Figma, REST APIs, Streamlit
- **Languages:** Proficient in English, French, and Chinese